# PG No. 43 $C_6(c)$ 6 [ hexagonal ]

* Wyckoff site: `1a`, site symmetry: `6..`

Table 1: Wyckoff bond: `1a@1a`, bond symmetry: `6..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, 0, Z]$ | $[0, 0, z]$ | `[1,2,3,4,5,6]` |

Table 2: Wyckoff bond: `3b@1a`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, 0]$ | $[0, 0, z]$ | `[1,-4]` |
| 2 | $[-Y, X-Y, 0]$ | $[0, 0, z]$ | `[2,-5]` |
| 3 | $[-X+Y, -X, 0]$ | $[0, 0, z]$ | `[3,-6]` |

Table 3: Wyckoff bond: `6c@1a`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $[0, 0, z]$ | `[1]` |
| 2 | $[-Y, X-Y, Z]$ | $[0, 0, z]$ | `[2]` |
| 3 | $[-X+Y, -X, Z]$ | $[0, 0, z]$ | `[3]` |
| 4 | $[-X, -Y, Z]$ | $[0, 0, z]$ | `[4]` |
| 5 | $[Y, -X+Y, Z]$ | $[0, 0, z]$ | `[5]` |
| 6 | $[X-Y, X, Z]$ | $[0, 0, z]$ | `[6]` |

* Wyckoff site: `6b`, site symmetry: `1`

Table 4: Wyckoff bond: `6a@6b`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $[x, y, z]$ | `[1]` |
| 2 | $[-Y, X-Y, Z]$ | $[-y, x-y, z]$ | `[2]` |
| 3 | $[-X+Y, -X, Z]$ | $[-x+y, -x, z]$ | `[3]` |
| 4 | $[-X, -Y, Z]$ | $[-x, -y, z]$ | `[4]` |
| 5 | $[Y, -X+Y, Z]$ | $[y, -x+y, z]$ | `[5]` |
| 6 | $[X-Y, X, Z]$ | $[x-y, x, z]$ | `[6]` |